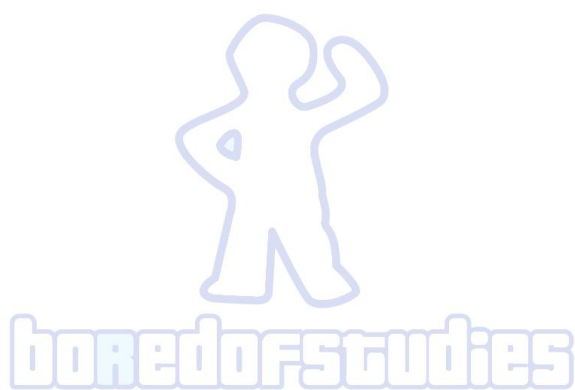




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2024

BORED OF STUDIES TRIAL EXAMINATION

6th October

Mathematics Extension 2

General instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using a black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks: 100**Section I – 10 marks** (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–14)

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Suppose that $\hat{a}$, $\hat{b}$ and $\hat{c}$ are unit vectors in three dimensional space.

What is the maximum value of $\hat{a} \cdot (\hat{b} + \hat{c})$?

- (A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

- 2 The velocity of a 2kg object at displacement x metres from the origin is given by $v = x^2$ metres per second.

What is the force acting on the particle in Newtons when $x = -2$?

- (A) -32 (B) -16 (C) 16 (D) 32

- 3 On the complex plane, the points A, B and C are represented by the complex numbers $-5 + 4i$, $1 + i$ and $-1 - 3i$ respectively.

Which of the following best describes $\triangle ABC$?

- (A) Equilateral triangle (B) Isosceles triangle
(C) Right-angled triangle (D) Scalene triangle

- 4 Which of the following statements is NOT always true?

- (A) $\forall x \in \mathbb{N} \quad \exists y \in \mathbb{N} \quad x + y \in \mathbb{N}$ (B) $\forall x \in \mathbb{N} \quad \exists y \in \mathbb{N} \quad \frac{x}{y} \in \mathbb{N}$
(C) $\forall x \in \mathbb{C} \quad \exists y \in \mathbb{C} \quad x = y^2$ (D) $\forall x \in \mathbb{C} \quad \exists y \in \mathbb{C} \quad xy = 1$

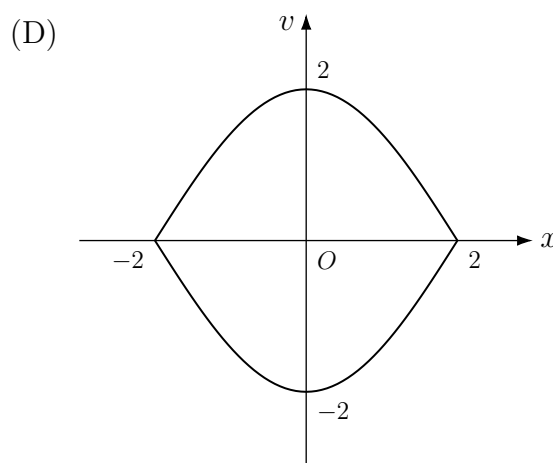
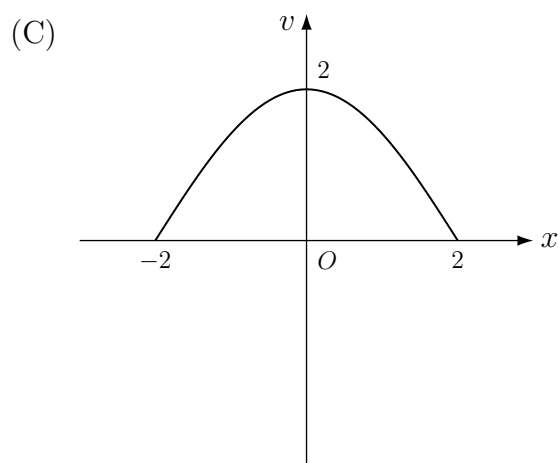
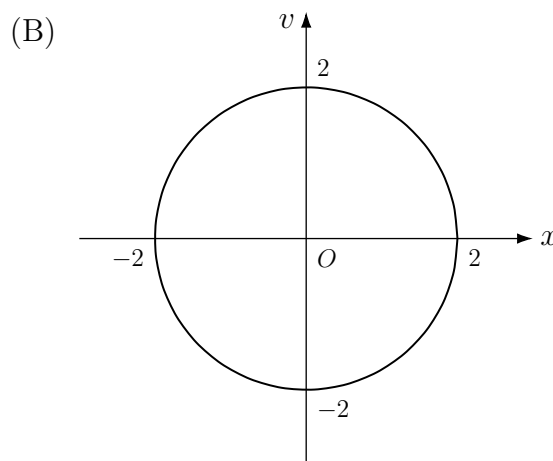
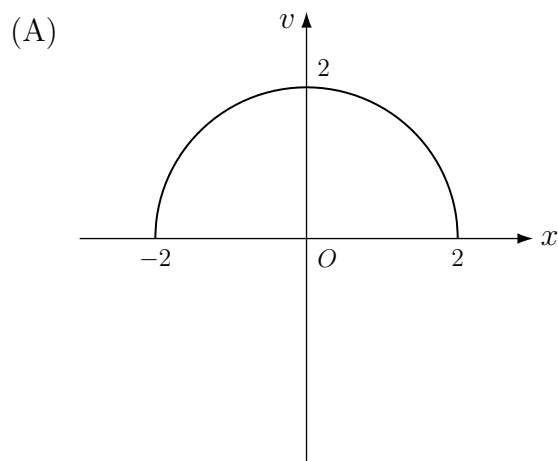
- 5 A quadratic polynomial $P(x) = ax^2 + bx + c$ has the roots z_1 and z_2 , where a is a positive real number, whilst b and c are non-zero complex numbers.

Let $\arg(z_1) = \alpha$, $\arg(z_2) = \beta$ and $\arg(z_1 + z_2) = \gamma$.

Which of the following is equal to $\arg(abc)$?

- (A) $\alpha + \beta + \gamma$ (B) $\alpha\beta\gamma$
 (C) $\alpha + \beta + \gamma + \pi$ (D) $\alpha\beta\gamma + \pi$
- 6 A particle moves in simple harmonic motion between $x = -2$ and $x = 2$ and has an acceleration equation of $\ddot{x} = -x$.

Which of the following best represents the graph of the particle's velocity versus displacement curve?



- 7** Let $\underline{a} = p\underline{i} + q\underline{j} + r\underline{k}$ where $0 < p < q < r$ and let $\underline{b}$ be a vector perpendicular to $\underline{a}$.

Suppose that $\underline{c}$ is a vector such that the angle between $\underline{a}$ and $\underline{b} - \underline{c}$ is obtuse.

Which of the following *could* be $\underline{c}$?

- (A) $-\underline{i} - \underline{j} - \underline{k}$ (B) $\underline{i} - \underline{j} + \underline{k}$ (C) $\underline{i} - \underline{j} - \underline{k}$ (D) $-\underline{i} + \underline{j} - \underline{k}$

- 8** Let α and β be any two non-real roots of the equation $z^6 = 1$.

Which of the following is always true?

- (A) $\operatorname{Re}(\alpha^2 + \beta^2) < 0$ (B) $\operatorname{Re}(\alpha^2 + \beta^2) > 0$
(C) $\operatorname{Re}(\alpha^2 - \beta^2) < 0$ (D) $\operatorname{Re}(\alpha^2 - \beta^2) > 0$

- 9** Which of the following is the value of $\int_0^1 \frac{4x - 1}{4x^4 - 4x^3 + x^2 + 1} dx$?

- (A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{4} - \ln 2$ (D) $\frac{\pi}{4} + \ln 2$

- 10** Let $P(z) = z^n \cos \theta_n + z^{n-1} \cos \theta_{n-1} + z^{n-2} \cos \theta_{n-2} + \cdots + z \cos \theta_1 + \cos \theta_0 - 2$, where $\theta \in \mathbb{R}$.

Which of the following is the smallest region on the complex plane that contains all the roots of $P(z)$?

- (A) $|z| < \frac{1}{3}$ (B) $|z| < \frac{1}{2}$ (C) $|z| > \frac{1}{3}$ (D) $|z| > \frac{1}{2}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

- (a) Evaluate $\int e^x \sec x \tan^2 x \, dx$. **3**
- (b) By considering the modulus-argument forms of $1 + i$ and $\sqrt{3} - i$, find the exact values of $\sin \frac{\pi}{12}$ and $\cos \frac{\pi}{12}$. **2**
- (c) The line ℓ_1 passes through the points $A(7, 6, 2)$ and $B(12, 10, 5)$. The line ℓ_2 passes through the points $C(1, -4, -8)$ and $D(11, 4, -2)$.
- (i) Show that ℓ_1 and ℓ_2 are parallel. **1**
- (ii) Find the vector equations of the line ℓ_1 using parameter λ and the line ℓ_2 using parameter μ . **2**
- (iii) Let P be a point on ℓ_2 . Find the length of the vector $\overrightarrow{AP}$ in terms of μ . **2**
- (iv) Hence, find the shortest distance between ℓ_1 and ℓ_2 . **2**
- (d) The roots of $P(x) = x^4 - 4x^3 + 6x^2 - 4x + 5$ are all non-real and make up the vertices of a square on the complex plane. Find the roots of $P(x)$. **3**

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

- (a) By considering an appropriate substitution, show that **2**

$$\int_0^{\pi} \frac{\sin^3 x}{\sec x + \tan x + \cos x} dx = 0.$$

- (b) Let z be a complex number which satisfies **3**

$$|e^z - 1| = 1 \quad \text{and} \quad 0 \leq \operatorname{Im}(z) < \frac{\pi}{2}.$$

Sketch the curve which describes the above relation on the complex plane.

- (c) Let $f(x)$ be a continuous and differentiable function over the interval $[a, b]$, where $a \neq b$.

- (i) Consider the following statement. **1**

If $f(a) < f(b)$ then there exists a $x_0 \in [a, b]$ where $f'(x_0) > 0$

By considering the contrapositive, explain why this statement is true.

- (ii) Consider the following statement. **1**

If $f(a) < f(b)$ then $f'(x) > 0$ for all $x \in [a, b]$

Show by counterexample that this statement is not always true.

Question 12 continues on page 7

Question 12 (continued)

- (d) Two particles P_1 and P_2 are oscillating with equal amplitudes along parallel number lines in simple harmonic motion. Let x_1 and x_2 be the displacement of the particles P_1 and P_2 respectively at time t . Their acceleration equations are

$$\ddot{x}_1 = -\frac{\pi^2}{4}x_1 \quad \text{and} \quad \ddot{x}_2 = -\frac{9\pi^2}{400}x_2.$$

Particle P_1 is initially at the midpoint between its centre of motion and its maximum displacement, whilst moving in the positive direction.

Particle P_2 is initially at the centre of its motion, whilst moving in the positive direction.

- (i) Find the time taken for P_1 to reach its maximum displacement for the fifth time. **2**
- (ii) When P_1 reaches its maximum displacement for the fifth time, particle P_2 also reaches its maximum displacement for the n th time. **2**

Find the value of n .

- (e) (i) By considering the complex roots of the equation $z^7 - 1 = 0$, show that the solutions to $x^3 + x^2 - 2x - 1 = 0$ are $2 \cos \frac{2\pi}{7}$, $2 \cos \frac{4\pi}{7}$ and $2 \cos \frac{6\pi}{7}$. **3**
- (ii) Hence, show that **1**

$$\sin\left(\frac{\pi}{14}\right) \sin\left(\frac{3\pi}{14}\right) \sin\left(\frac{5\pi}{14}\right) = \frac{1}{8}.$$

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

- (a) Suppose that z_1 and z_2 both have a modulus of 1. Show that **1**

$$|1 + z_1| + |1 + z_2| + |1 + z_1 z_2| \geq 2.$$

- (b) An object of mass m is projected vertically upwards from the ground. At time t the object has a velocity of v and a displacement of y relative to the ground. The object is initially at the origin with a velocity of u .

The object is subjected to a gravitational force of magnitude mg and a resistance force of magnitude mkv^2 where g is the acceleration due to gravity and k is a positive constant. Let upwards be the positive direction for the equations of motion.

- (i) Explain why the object's acceleration equation for both upwards and downwards motion can be written as **1**

$$\ddot{y} = -g - kv|v|.$$

- (ii) Let H be the maximum height of the object's motion. Show that **2**

$$H = \frac{1}{2k} \ln \left(\frac{g + ku^2}{g} \right).$$

- (iii) Let T_u be the time taken for the object to reach its maximum height. Show that **2**

$$T_u = \frac{1}{\sqrt{kg}} \tan^{-1} \left(u \sqrt{\frac{k}{g}} \right).$$

- (iv) After reaching its maximum height, the object then falls back to its initial position on the ground at a velocity of v_g . Show that **3**

$$v_g^2 = \frac{gu^2}{g + ku^2}.$$

- (v) Let T_d be the time taken for the object to fall from its maximum height back to the ground. Show that **4**

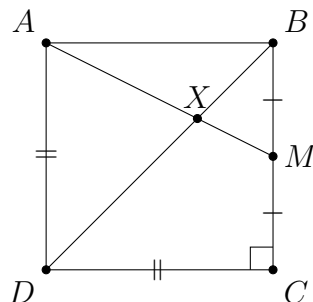
$$T_d = \frac{1}{\sqrt{kg}} \ln \left(\frac{\sqrt{g + ku^2} + u\sqrt{k}}{\sqrt{g}} \right).$$

- (vi) Hence, show that $T_d > T_u$. **2**

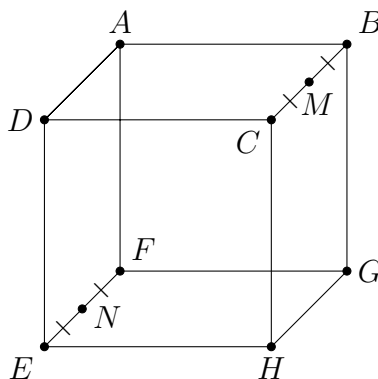
End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet

- (a) Suppose that $ABCD$ is a square. Let M be the midpoint of BC and X be the intersection of BD and AM , as shown in the diagram below.



- (i) Let $\overrightarrow{AB} = u$ and $\overrightarrow{AD} = v$. Show that $\overrightarrow{AX} = \frac{2}{3}u + \frac{1}{3}v$. **2**
- (ii) Hence, show that $|\overrightarrow{XD}| = 2|\overrightarrow{BX}|$. **1**
- (iii) Let P be the midpoint of AB . By considering the vector equation of the line in the direction of $\overrightarrow{PC}$, show that X lies on PC . **2**
- (b) A cube with side length of 2 units has vertices A, B, C, D, E, F, G and H as shown in the diagram below. Let M and N be the midpoints of BC and EF respectively.



- (i) Let $\overrightarrow{AB} = u$, $\overrightarrow{AD} = v$ and $\overrightarrow{AF} = w$. By expressing the sides in terms of u, v and w , prove that $AMHN$ is a rhombus. **2**
- (ii) Let P and Q be the midpoints of AB and EH respectively. Suppose that PC intersects AM at point X and NH intersects FQ at Y . **2**

Using the result(s) in part (a), show that $|\overrightarrow{XY}| = \frac{2\sqrt{11}}{3}$.

Question 14 continues on page 10

Question 14 (continued)

(c) For integer $n \geq 0$ and $a > 0$, let $I_n = \int_0^1 \frac{x^n}{x+a} dx$.

(i) Show that for integer $n \geq 1$ **1**

$$I_n = \frac{1}{n} - aI_{n-1}.$$

(ii) Deduce that **2**

$$I_n = \sum_{k=0}^{n-1} \frac{(-1)^k a^k}{n-k} + (-1)^n a^n \ln \left(1 + \frac{1}{a} \right).$$

(iii) Hence, find the value of $\int_0^1 \frac{x^4}{(x+1)(2x+1)} dx$. **3**

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet

- (a) Suppose that the point $(0, 0, -1)$ lies on the plane $\mathcal{P}$ in three dimensional space. The position vector $\underline{i} - \underline{k}$ is perpendicular to this plane.

(i) By considering a general point $P(x, y, z)$ on the plane $\mathcal{P}$ and an appropriate dot product, show that the Cartesian equation of the plane is $x - z = 1$. **1**

(ii) A cylinder with Cartesian equation $x^2 + y^2 = 8$ for all values of z intersects with the plane $\mathcal{P}$. The set of all points of intersection describes the curve $\mathcal{C}$ in three dimensional space. **1**

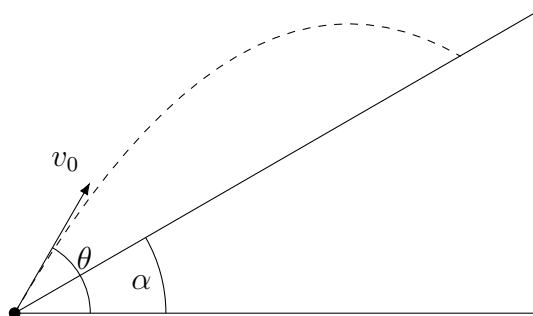
By considering the parametric coordinates of a circle, find the coordinates of a general point $Q(x(\theta), y(\theta), z(\theta))$ on $\mathcal{C}$, in terms of parameter θ .

(iii) It can be shown that at a given point of contact $R(x(\theta_c), y(\theta_c), z(\theta_c))$, the direction vector of the tangent line $\underline{r}_d$ to the curve $\mathcal{C}$ is **3**

$$\underline{r}_d = x'(\theta_c)\underline{i} + y'(\theta_c)\underline{j} + z'(\theta_c)\underline{k}. \quad \text{(Do NOT prove this)}$$

Suppose that the line ℓ is tangent to the curve $\mathcal{C}$ with a point of contact of $(2, 2, 1)$. Find the vector equation of the line ℓ .

- (b) An object is projected at the bottom of a frictionless inclined plane with an initial speed of v_0 and an acute angle of θ to the horizontal. The inclined plane has an acute angle of α to the horizontal, where $\alpha < \theta$. **2**



Let g be the acceleration due to gravity. The displacement vector of the object is

$$\underline{r} = v_0 t \cos \theta \underline{i} + \left(v_0 t \sin \theta - \frac{gt^2}{2} \right) \underline{j}. \quad \text{(Do NOT prove this)}$$

Let T be the time taken for the object to land on the inclined plane. Show that

$$T = \frac{2v_0 \cos \theta}{g} (\tan \theta - \tan \alpha).$$

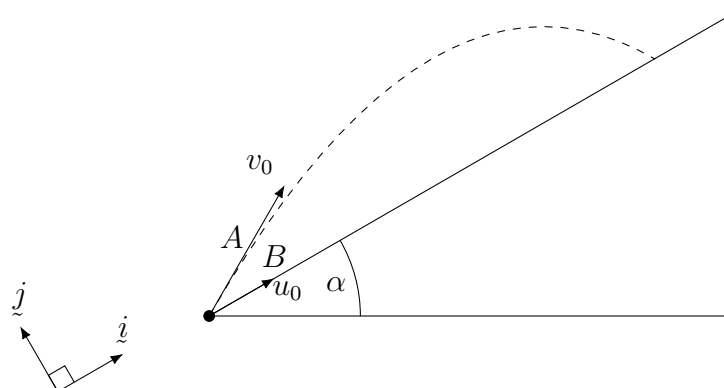
Question 15 continues on page 12

Question 15 (continued)

- (c) Object A is projected at the bottom of a frictionless inclined plane with an initial speed of v_0 and an acute angle of θ to the horizontal. The inclined plane has an acute angle of α to the horizontal, where $\alpha < \theta$.

At the same time, another object B is launched uphill along the plane with an initial speed of u_0 .

Let $\hat{i}$ and $\hat{j}$ be unit vectors parallel and perpendicular to the inclined plane respectively, as shown in the diagram below. The only force action acting on both objects is gravity with acceleration g .



Let $\underline{r}_A$ and $\underline{r}_B$ be the displacement of the objects A and B respectively at time t .

- (i) By finding the acceleration equation of A in terms of g and α , show that **3**

$$\underline{r}_A = \left(v_0 t \cos(\theta - \alpha) - \frac{gt^2 \sin \alpha}{2} \right) \hat{i} + \left(v_0 t \sin(\theta - \alpha) - \frac{gt^2 \cos \alpha}{2} \right) \hat{j}.$$

- (ii) Find $\underline{r}_B$ in terms of g, t, u_0 and α . **2**

- (iii) Suppose that the two objects collide when object B reaches its maximum displacement along the plane. **3**

Use the result in part (b) to show that

$$\tan(\theta - \alpha) \tan \alpha = \frac{1}{2}.$$

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet

- (a) For positive integer n and $a_k \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ with $a_{2n-1} \neq 0$, let

$$u_{2n} = \sum_{k=0}^{2n-1} 10^k a_k \quad \text{and} \quad w_{2n} = \sum_{k=0}^{2n-1} (-1)^k a_k.$$

Note that u_{2n} represents a $2n$ digit number with leading digit a_{2n-1} and final digit a_0 .

- (i) Suppose that w_{2n} is divisible by 11 for all positive integer n . Prove by mathematical induction that u_{2n} is also divisible by 11. **3**
- (ii) A *palindromic* number is a number which is unchanged when it's digits are in reverse order. Using the result in part (i), deduce that all palindromic numbers with an even number of digits is divisible by 11. **1**
- (b) Let $f(x)$ be a continuous function where $f(x) > 0$ and $f'(x) > 0$ over the domain $[a, b]$. Let $F(x)$ be a function where $F'(x) = f(x)$. **2**

Show that

$$\int_a^b \frac{f(x)}{f'(x)} dx \geq 2[F(b) - F(a)] + \frac{1}{2}[f(a) - f(b)][f(a) + f(b)].$$

Question 16 continues on page 14

Question 16 (continued)

- (c) Suppose that $\pi = \frac{p}{q}$ where p and q are positive integers with no common factors other than 1.

For integer $n > 1$, let

$$f(x) = \frac{x^n(p - qx)^n}{n!}$$

and

$$g(x) = f(x) + \sum_{k=1}^n (-1)^k f^{(2k)}(x),$$

where $f^{(2k)}(x)$ is the $2k$ th derivative of $f(x)$. For example, $f^{(2)}(x)$ is the second derivative of $f(x)$ and $f^{(4)}(x)$ is the fourth derivative of $f(x)$.

- (i) Show that $f(\pi - x) = f(x)$. 1

- (ii) Explain why $g(0) + g(\pi)$ is an integer. 3

- (iii) By evaluating $\frac{d}{dx} [g'(x) \sin x - g(x) \cos x]$ show that $\int_0^\pi f(x) \sin x \, dx$ is an integer. 2

- (iv) Show that 1

$$0 \leq \int_0^\pi f(x) \sin x \, dx \leq \frac{\pi^{n+1} p^n}{n!}.$$

- (v) Suppose that n is sufficiently large such that there exists a positive integer m where $\pi p < m < n$. Show that 1

$$\frac{(\pi p)^n}{n!} < \frac{(\pi p)^m}{(m-1)!} \cdot \frac{1}{n}.$$

- (vi) Hence, deduce that π is irrational. 1

End of paper

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